

CS/E - 180

RevolutionParts Inc.: Quoting

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Project Overview:

- ❖ Create an effective quoting functionality for shipments that replaces the existing quoting service—EasyPost—in RP's codebase
- ❖ Create an effective caching mechanism that reduces API calls.

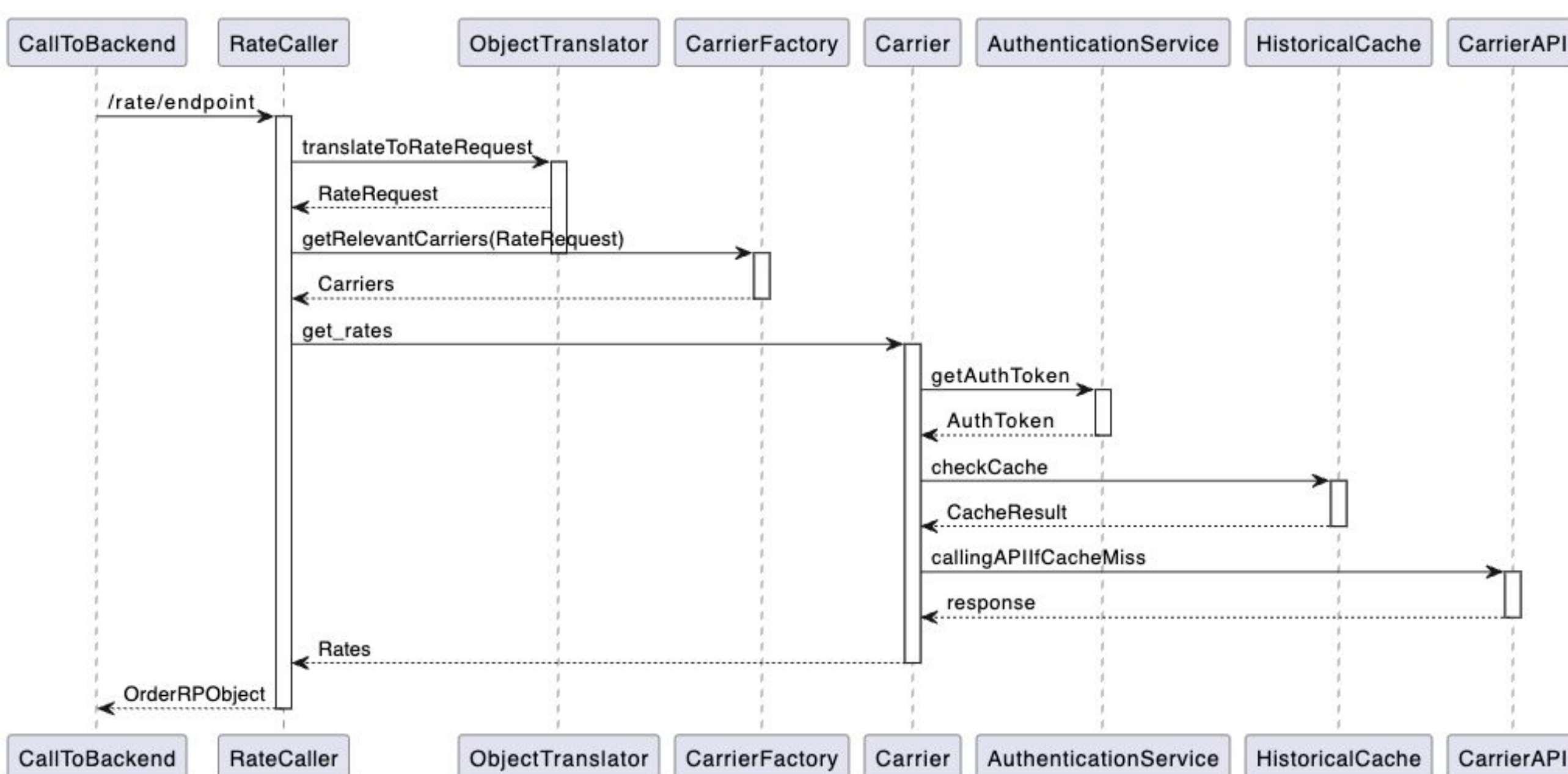
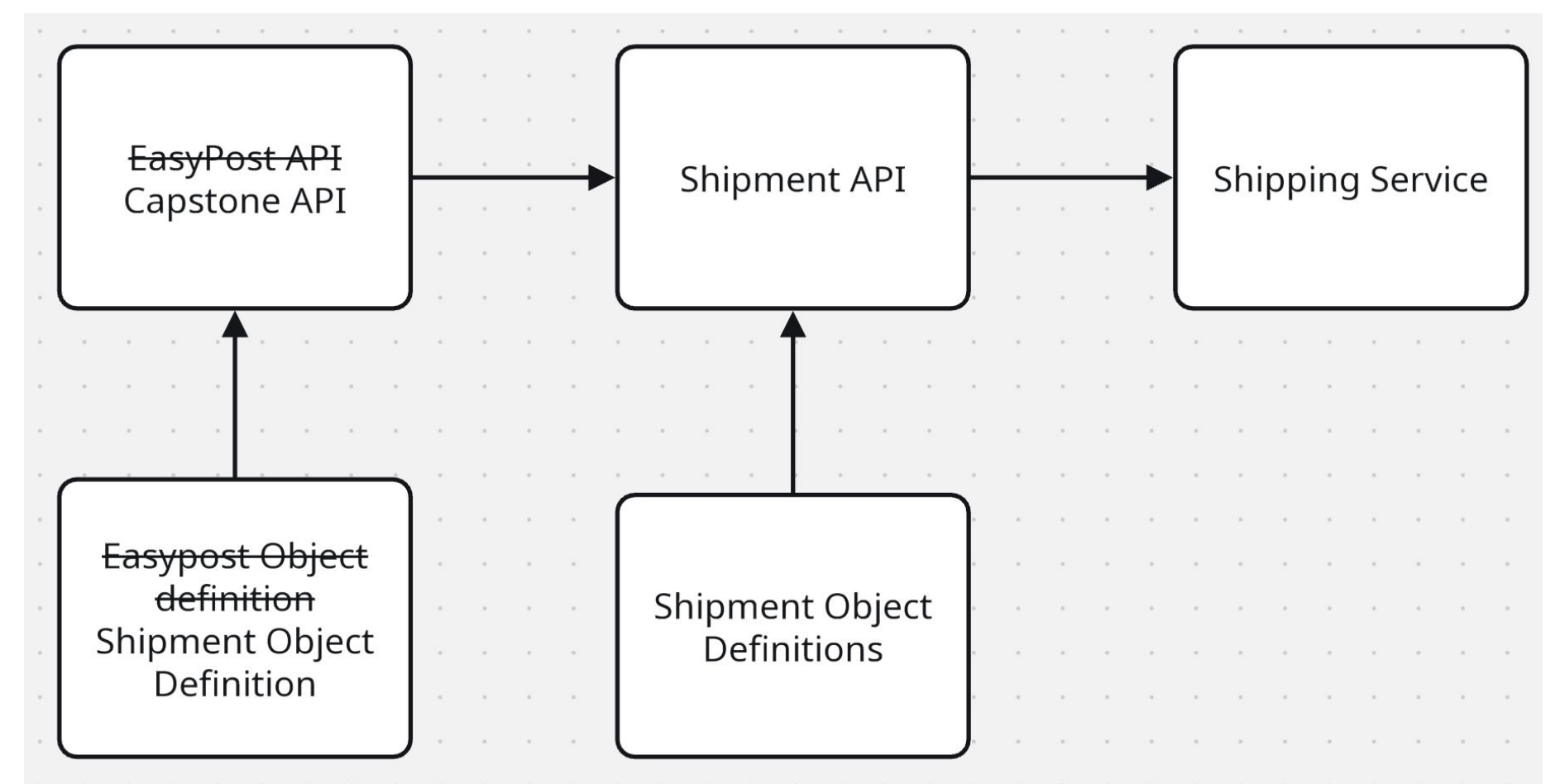
Design Description/Justification:

❖ Cache Creation :

- We created a historical and refreshable cache that takes in data from a CSV file and loads into in PostgreSQL server and is queryable over HTTP.

EPIC: Customer Archetypes:

- ❖ Quoting
- ❖ Caching
- ❖ Autopart Buyers
- ❖ Autopart Suppliers
- ❖ Inventory Managers



Results:

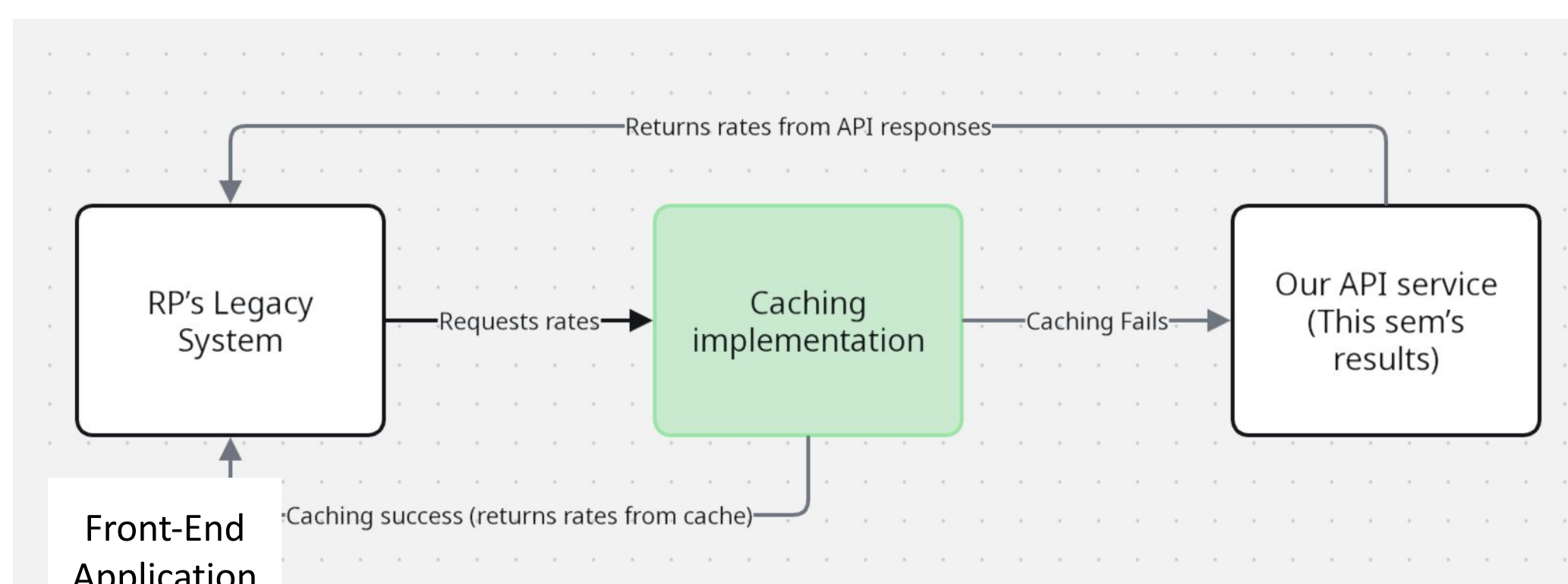
- ❖ Developed an intuitive interface tool to view different rates returned from API carriers based on package description
- ❖ Designed a backend server with API endpoints that forwards requests to the cache and then to carrier APIs if the cache missed
- ❖ Successfully retrieves address and data response from historical data/carrier APIs along with side-by-side delay times for cache hits and misses
- ❖ Implemented cache system for quicker responses, and less reliance on third-party API services.
 - More time to get a quote means less sales.
 - Reduces cost of calling external APIs.

Value for RP:

- ❖ Financial
- ❖ Customer Satisfaction

Lessons Learned:

- ❖ The importance of data analysis and problem understanding for caching
- ❖ Realistic expectations of project application in real world scenarios



Future Work:

- ❖ Integrate API with RevolutionParts' code base.
- ❖ Gain API credentials for UPS.
- ❖ Gain production API credentials for all carriers.

The screenshot shows a web application for generating shipping quotes. It includes sections for 'Ship From' (Full Name, Street Address, City, State, Zipcode), 'Package Details' (Dimensions, Weight, Package Type, Signature Required), 'Ship To' (Full Name, Street Address, City, State, Zipcode), 'Select Carriers' (UPS, FedEx, USPS), and 'Rates' (USPS Connect Local, USPS Ground Advantage, USPS Priority Mail, USPS Priority Mail Express). The 'Rates' section shows a table of rates with columns for carrier, service, and price. The 'USPS Ground Advantage' rate is highlighted with a blue border and a price of \$6.43. The 'Cache Status' section shows a green 'HIT' indicator and a 'Historical Median Price' of \$6.43.

